

1. Administrative & Study Identification

Official Study Title: A Phase 3, Open-label Study of Ifinatamab Deruxtecan Versus Docetaxel in Participants With Metastatic Castration-Resistant Prostate Cancer (mCRPC) (IDeate-Prostate01)
Public Title: A Clinical Study of Ifinatamab Deruxtecan (I-DXd) in People With Metastatic Prostate Cancer (MK-2400-001) **Short Title/Acronym:** IDeate-Prostate01 **Protocol Number:** MK-2400-001 **NCT Number:** NCT06925737 **Trial Status:** RECRUITING **Sponsor/Company Name:** Merck (Merck Sharp & Dohme LLC in collaboration with Daiichi Sankyo) **Investigational Product:** Ifinatamab Deruxtecan (I-DXd / MK-2400) **Phase of Development:** Phase III **Medical Monitor:** Senior Vice President and Head of Oncology, Global Clinical Development, Merck Research Laboratories

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To evaluate if patients with mCRPC who receive ifinatamab deruxtecan (I-DXd) live longer overall (OS) and live longer without the cancer growing or spreading (radiographic progression-free survival, rPFS) compared to those who receive standard-of-care chemotherapy (docetaxel). **Secondary Objectives:** To assess the objective response (OR), duration of response (DOR), time to first subsequent therapy (TFST), and evaluate the safety and tolerability of the investigational drug.

2.2 Background & Rationale To address critical unmet medical needs for patients with metastatic castration-resistant prostate cancer (mCRPC) whose tumors progressed following prior treatment with an androgen receptor pathway inhibitor (ARPI). This study marks a pivotal trial for a potential first-in-class B7-H3 directed antibody-drug conjugate (ADC).

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Active Comparator (Docetaxel) **Blinding:** Open-label **Randomization:** 1:1, Stratified (by metastatic pattern [liver vs. bone-only vs. other], geographic region, B7-H3 expression level, and prior receipt of PSMA-targeted radionuclide therapy)

3.2 Planned Enrollment & Duration Target **SN\$:** 1,440 patients **Number of Sites:** 216 Global Locations **Estimated Study Start:** 07-Apr-2025 **Estimated Primary Completion:** Jun-2028

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Male, Age ≥ 18 years. 2. Has diagnosis of histologically or cytologically confirmed metastatic castration-resistant prostate cancer (mCRPC). 3. Has prostate cancer progression while on androgen deprivation therapy (ADT) (or post bilateral orchiectomy) within 6 months prior to entering the study. 4. Has received prior treatment with 1 or 2 androgen receptor pathway inhibitors (ARPI) and progressed during or after at least 8 weeks of treatment.

4.2 Exclusion Criteria 1. History of (non-infectious) interstitial lung disease (ILD)/pneumonitis that required steroids. 2. Has uncontrolled or significant cardiovascular disease. 3. Has received prior treatment with a taxane-based chemotherapy agent for mCRPC.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: Ifinatamab deruxtecan (I-DXd) administered via IV (Active comparator receives Docetaxel). **Route of Administration:** Intravenous (IV). **Duration of Treatment:** Until disease progression, unacceptable toxicity, or withdrawal.

5.2 Concomitant & Prohibited Therapy Permitted: Prednisone (with docetaxel) and continued androgen deprivation therapy (ADT). **Prohibited:** Taxane-based chemotherapy agent for mCRPC prior to the study.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening	Day -28 to 0	Informed Consent, Medical History, Tumor Imaging/Scans, Safety Labs
Baseline	Day 0	Randomization, First Infusion Dose
Interim	Infusion Cycles	Safety Labs, Efficacy Assessment (Radiologic scans), AE Monitoring
End of Study	Follow-up	Final Vital Signs, Treatment Discontinuation Interview, Survival Follow-up

7. Outcome Measures (Endpoints)

7.1 Primary Endpoints * **Overall Survival (OS):** Defined as the time from randomization to death due to any cause. * **Radiographic Progression-Free Survival (rPFS):** Defined as the time from randomization to the first documented disease progression per prostate cancer working group (PCWG)-modified Response Evaluation Criteria In Solid Tumors (RECIST) 1.1 by blinded independent central review (BICR) or death due to any cause, whichever occurs first.

7.2 Secondary & Exploratory Endpoints * Time to First Subsequent Therapy (TFST) * Objective Response (OR) * Duration of Response (DOR) * Safety parameters and occurrence of treatment-related adverse events

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Continuous evaluation via HCP monitoring, particularly watching for respiratory symptoms suggestive of ILD/pneumonitis. **Serious Adverse Events (SAEs):** Serious adverse events require reporting typically within 24 hours of site awareness. **Data Monitoring Committee (DMC):** Independent safety monitoring committee.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Intent-to-Treat (ITT) population for primary efficacy analysis; Safety Set for AE evaluations. **Sample Size Justification:** \$N = 1,440\$ patients powered to evaluate the superiority of ifinatumab deruxtecan vs. docetaxel. **Handling of Missing Data:** Standard oncology guidelines; censoring at the last known alive date or last evaluable progression-free radiologic assessment.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki. **Monitoring Plan:** Frequency of onsite and remote monitoring visits governed by the sponsor's clinical operations plan. **Audit Trail:** Validated eCRF locking and independent data clarification mechanisms.

11. Study Identifiers & Governance

Primary ID: MK-2400-001 **Registry Number:** NCT06925737 **Internal Posting ID:** 353553842076402844 **Sponsor/Lead Organization:** Merck **Study Title:** IDEate-Prostate01 **Official Regulatory Title:** A Phase 3, Open-label Study of Ifinatumab Deruxtecan Versus Docetaxel in Participants With Metastatic Castration-Resistant Prostate Cancer (mCRPC) (IDEate-Prostate01)

12. Clinical Framework (Oncology Specific)

Primary Condition / Preferred Name: Metastatic Prostate Cancer (mCRPC) **Disease Areas:** Prostate Cancer, Prostatic Neoplasms, Neoplasms (Neoplastic Process) **MeSH Code:** D009369 **Diagnosis**

Threshold: Prostate cancer progression while on ADT within 6 months prior **Medical Methodology:** B7-H3 directed Antibody-Drug Conjugate (ADC) vs. Chemotherapy **Optimization Target:** Improved OS and rPFS

13. Study Procedures & Methodology

Management Pathway: Oncology / Infusion treatment pathway
Education Protocol (HCP): Investigator protocol training and ILD/ pneumonitis symptom awareness **Education Protocol (Patient):** Informed consent onboarding and adverse event self-reporting
Quality Audit Mechanism: Clinical Research Associate (CRA) source data verification

14. Observation & Follow-Up Schedule

Total Duration: Primary completion anticipated by June 2028
Onsite Visit Cadence: Aligned with infusion cycles and radiologic imaging schedules **Remote Monitoring Frequency:** Continuous post-treatment survival and subsequent therapy tracking **Checklist Review Interval:** Evaluated periodically prior to each sequential dosing cycle

15. Sign-off & Approval

Role	Name	Signature	Date	:--	:--	:--	:--
Principal Investigator	[Name]	____	[DD-MMM-YYYY]				
Clinical Operations Lead	[Name]	____	[DD-MMM-YYYY]		Lead		
Statistician	[Name]	____	[DD-MMM-YYYY]				